

The Modern Rise of STIs in the U.S.

Exploring Long-Term Syphilis Trends and Regional Inequities

Group AA-3: Yan Peng, Mavis Yu, Ting-Yu Hsu,
Houzhengwu Shi



What is STI? Why do we care STIs?

What is STI?

Sexually Transmitted Infections (STIs) are infections passed through sexual contact. Common STIs include gonorrhea, syphilis, chlamydia, and HIV. They often have no symptoms but can cause serious health problems if untreated.

Why do we care?

- **STIs are rising fast** — especially among young people and marginalized communities.
- **Many go undiagnosed** — some STIs show no symptoms but can still spread and lead to infertility, pregnancy complications, or chronic pain.
- **Unequal impact** — infection rates are much higher in certain regions or populations, often due to poverty, limited healthcare access, and education gaps.
- **Our goal** is to help **policymakers and public health planners** understand where and why STI like syphilis is most concentrated, so they can design better prevention programs.

This issue affects **individual health, public health systems, and health equity nationwide.**

Overview of our Project

Our analysis focuses on the following four questions:

1. How have syphilis rates changed nationwide from 2000 to 2023?
2. How does syphilis compare to other STIs (gonorrhea and chlamydia) in terms of trends and severity from 2000 to 2023?
3. Do states with historically rising syphilis trends also exhibit high infection rates among women aged 15–44 in 2023?
4. Which states and regions should be prioritized for future funding and interventions?

Overview of Data

Data set 1: AtlasPlusTableData

We selected the AtlasPlus table data for analysis. It contains demographic and health-related variables (e.g., population counts, socioeconomic indicators, and other county-level metrics) that can help us explore relationships between regional characteristics and health outcomes.

Data set 2: PS-Syphilis-Rates-Women-15-44-Years-by-County-US-2023

We chose the 2023 syphilis-rates dataset for women aged 15 – 44, broken down by U.S. county. This will allow us to map and analyze the geographic distribution of syphilis incidence among women of childbearing age.

All data was accessed in May 2025 and is publicly available. Both datasets are collected and published by the CDC as part of its national STI surveillance system.

Number of observations in Atlas dataset: 8208
Total missing values in Atlas dataset: 1347

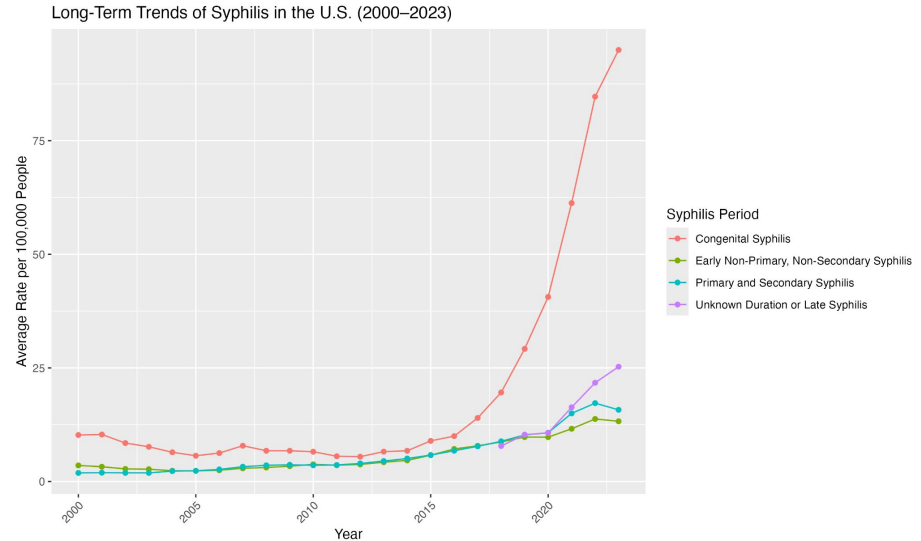
Number of observations in PS dataset: 3144
Total missing values in PS dataset: 9

How have syphilis rates changed nationwide from 2000 to 2023?

Four periods of syphilis:

- (1) **Congenital Syphilis**: passed from an infected pregnant person to their baby.
- (2) **Primary and Secondary Syphilis**: The early stages of syphilis infection.
- (3) **Early Non-Primary, Non-Secondary Syphilis**: A stage where the infection is recent but without symptoms.
- (4) **Unknown Duration or Late Syphilis**: present over a year with no symptoms.

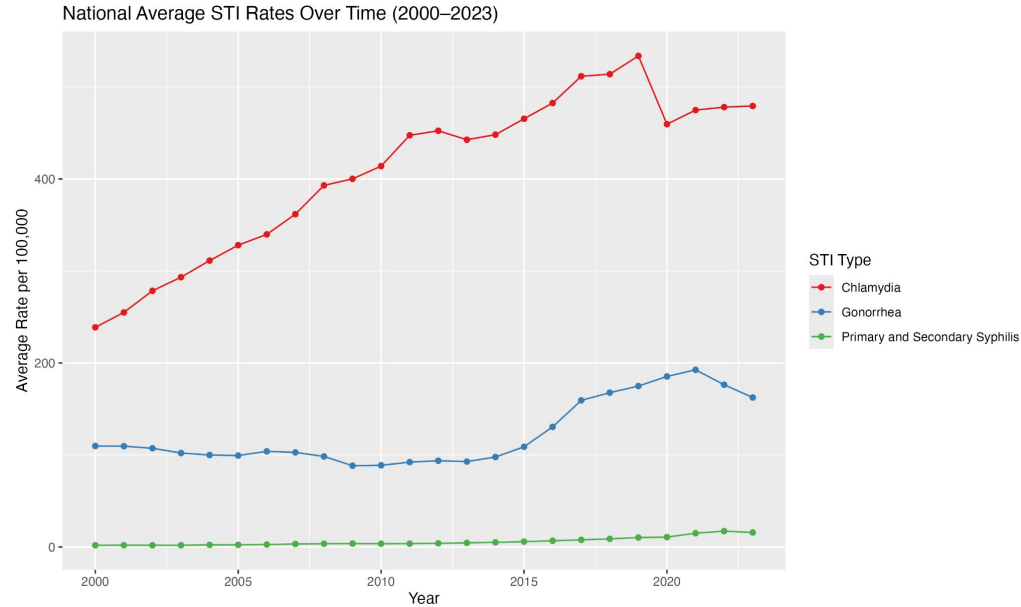
- For most of the **early 2000s**, all types of syphilis remained relatively stable and low.
- Starting around **2015**, all types began to increase noticeably.
- The most alarming rise is in congenital syphilis, which surged exponentially after **2019**, reaching nearly 100 per 100,000 people by 2023.



We focus on primary and secondary syphilis in this project!

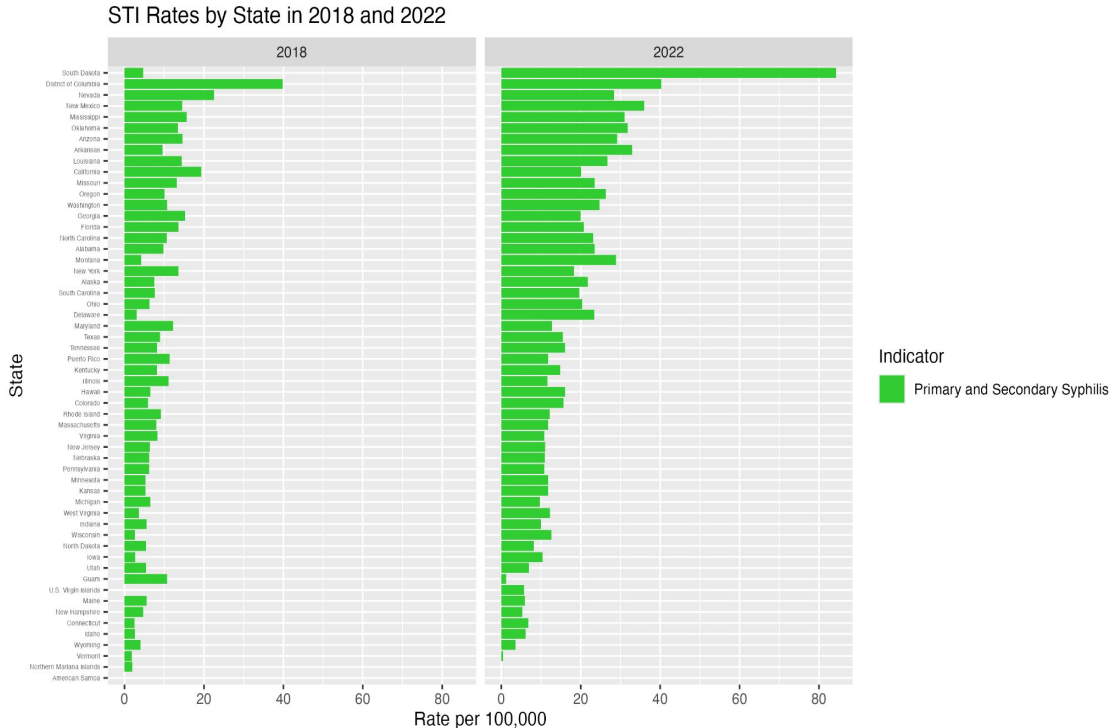
How does syphilis compare to other STIs (gonorrhea and chlamydia) in terms of trends and severity from 2000 to 2023?

- All three STIs show a long-term upward trend.
- **Chlamydia** consistently shows the highest infection rate, nearly doubling from ~250 to over 450 per 100,000 people.
- **Gonorrhea** remained relatively stable until 2015, peaking around 2021.
- **Primary and secondary syphilis** increased steadily and nearly tripled from 2000 to 2023.



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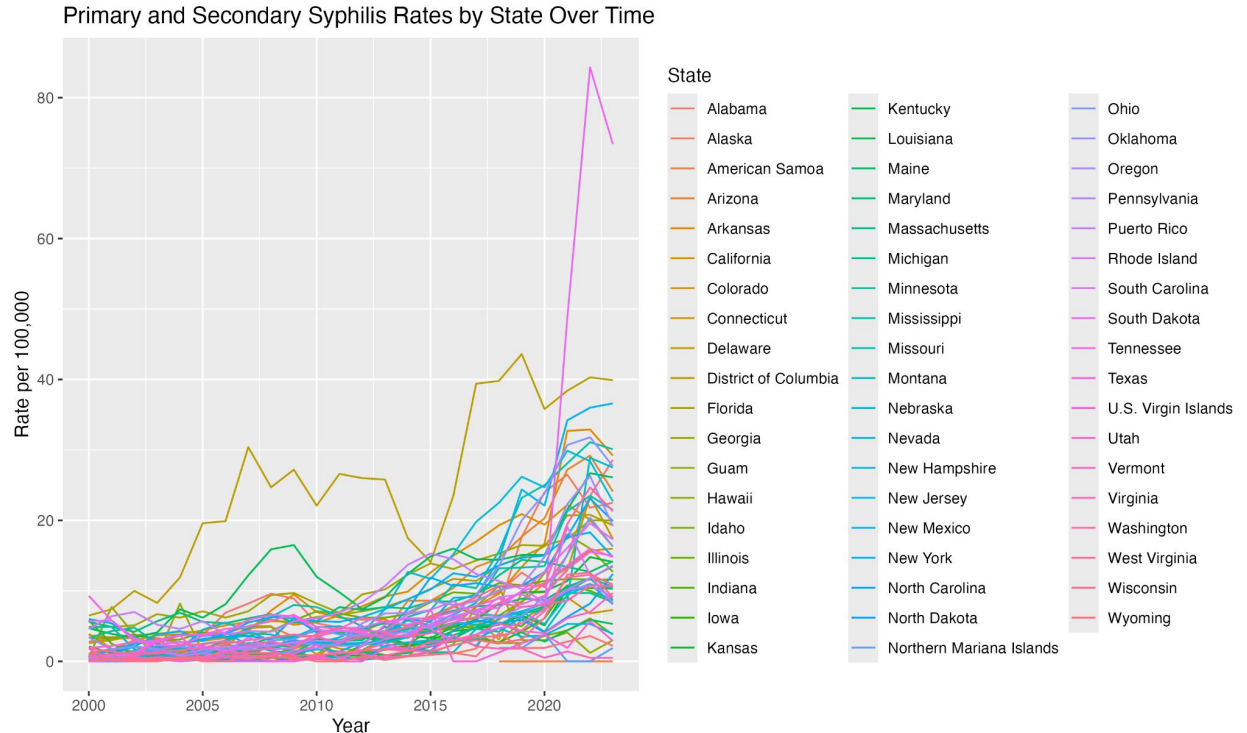
- A nationwide increase in primary and secondary syphilis rates.
- States like South Dakota, District of Columbia, and Mississippi had high STI rates for syphilis.
- The rapid and consistent rise over time makes it a critical concern for early intervention and awareness.



03

Do states with historically rising syphilis trends also exhibit high infection rates among women aged 15–44 in 2023?

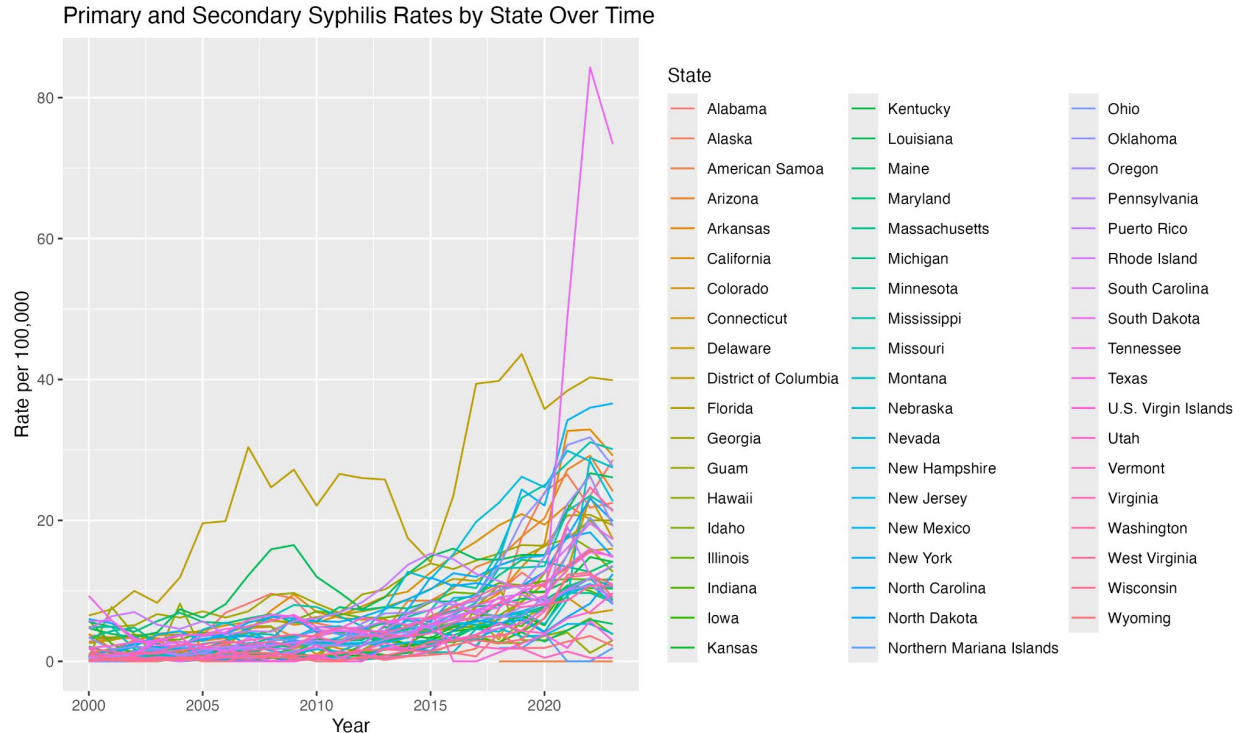
- **2000–2013:** Most states experienced low and steady syphilis rates.
- **2014–2019:** Several states began showing a gradual upward trend, indicating the start of a resurgence.
- **2020–2023:** A sharp and widespread increase occurred, with some states more dramatically affected.



03

Do states with historically rising syphilis trends also exhibit high infection rates among women aged 15–44 in 2023?

- **South Dakota:** Exhibited the most pronounced surge post-2020, the highest nationally.
- **District of Columbia and Louisiana:** Had already elevated rates in earlier years, reflecting consistently high burdens.
- **New Mexico** also showed persistently elevated trends.

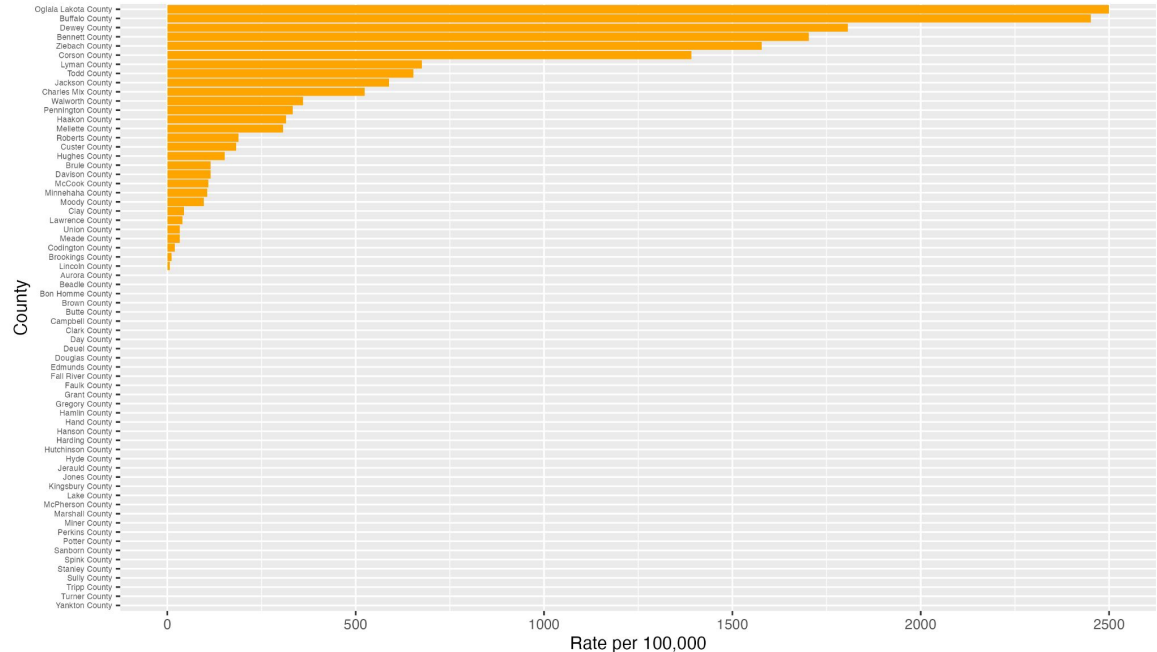


04

Which states and regions should be prioritized for future funding and interventions?

- There is a huge disparity in infection rates across counties in South Dakota.
- **Oglala Lakota County** has the highest rate: over 2,400 per 100,000, far exceeding all other counties.
- **Buffalo, Dewey, and Bennett** show alarmingly high rates, suggesting localized outbreaks.

Primary and Secondary Syphilis Rates by County in South Dakota (2023)



04

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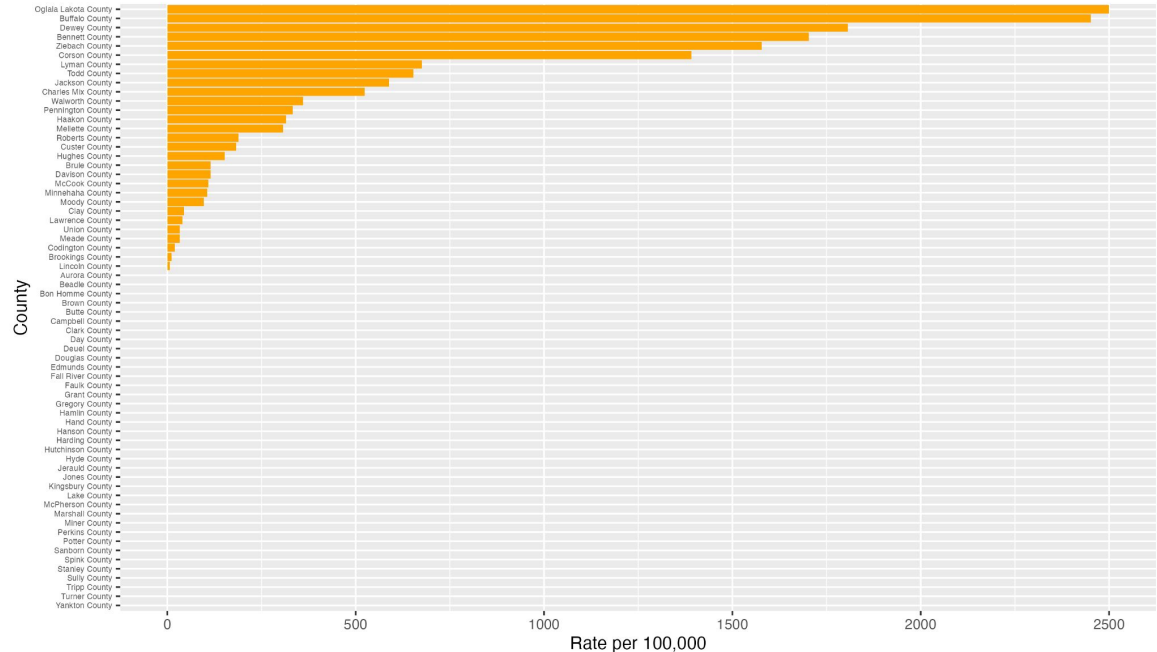
- **Rate = (Cases / Population) × 100,000**

A small number of cases in a very small population will produce a very large rate.

- **Oglala Lakota County**
Population: ~13,000.

- **Buffalo County:**
Indian Reservation.
One of the least populated counties in the U.S., ~2,000 people.

Primary and Secondary Syphilis Rates by County in South Dakota (2023)



Over half the counties report **zero cases**, showing a strong geographic concentration of the problem.

Conclusion

01 How have syphilis rates changed nationwide from 2000 to 2023?

Syphilis rates stayed low and steady through most of the 2000s, then began climbing around 2015. While all stages rose, congenital syphilis exploded after 2019, reaching nearly 100 cases per 100,000 by 2023. Primary and secondary syphilis have also trended upward, so focusing on those stages is key for understanding recent increases.

02 How does syphilis compare to other STIs (gonorrhea and chlamydia) in terms of trends and severity from 2000 to 2023?

Syphilis has increased much more quickly than gonorrhea or chlamydia since 2015. It started in a few states in 2018 and spread to many states by 2022.

Conclusion

03 Do states with historically rising syphilis trends also exhibit high infection rates among women aged 15–44 in 2023?

From 2000 to 2013, syphilis rates stayed low and steady in most states. But starting around 2014, rates began to rise, and the increase became much sharper after 2020. Some states with rising syphilis trends, like South Dakota, also show very high 2023 infection rates.

04 Which states and regions should be prioritized for future funding and interventions?

Oglala Lakota and Buffalo counties in South Dakota have very high syphilis rates, even though their populations are small. This means they could be top priorities for help, but we need more data on case numbers to be sure.

Future Plan

- Merge county-level socioeconomic data (e.g., poverty and healthcare access) to identify the drivers of high syphilis rates.
- Collect accurate raw case numbers from as many regions and states as possible to avoid low population bias.
- Produce promotional posters or short videos to raise public awareness of syphilis prevention and control.

Thank you !



Q&A

